

Open-Source Cross-Domain Datasets Enabled by LLM-Driven Digital Twins and Network Telemetry

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Machine-learning-enabled network research requires both practical telemetry and controllable synthetic data, but optical transport, fibre sensing, and wireless access data are usually collected and released separately. We present a cross-domain dataset framework with two complementary parts: practical datasets organised through a common telemetry and metadata structure, and generated datasets produced through an LLM-driven multi-agent digital-twin (DT) workflow. “Cross-domain” denotes common representation, repository, and task-level fusion across domains; the released practical campaigns were acquired independently and are not contemporaneous end-to-end measurements. We evaluate a constructed optical–wireless interoperability benchmark using a leakage-free chronological split. A three-dimensional privacy-aware wireless representation achieves a binary service-risk Macro-F1 of 0.646 ± 0.005 over ten seeds, compared with 0.646 ± 0.004 for a hand-crafted representation and 0.633 ± 0.004 for raw wireless features. Its strongest observed CPE-attribute attack reaches Macro-F1 0.238 ± 0.011 , compared with 0.276 on raw telemetry; PCA exposes less CPE information but also reduces utility. For generated data, a corrected GNPpy feasibility demonstration contains 8,192 execution records but 6,144 unique configurations, exactly matching an independently specified deterministic enumeration. A separately identified new reproduction retains six complete LLM request/response pairs; its manual recovery and subsequent deterministic replay are disclosed. A recovered-source audit reproduces 96 stratified archived results with a pinned GNPpy v2.12 compatibility environment to within 0.0005 dB; a schema-corrected GNPpy 2.14.2 check executes all 96 but differs in GSNR by 0.628 dB on average and retains configuration warnings. These results position the LLM workflow as a feasibility demonstration and the semantic module as a measured privacy–utility trade-off rather than a privacy guarantee.

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1. INTRODUCTION

In recent years, machine learning (ML) has been widely investigated in telecommunication networks to support intelligent and autonomous network operation. In optical networks, ML has been applied to quality of transmission (QoT) estimation, traffic prediction, anomaly detection, and resource allocation [1]. In wireless networks, ML-based methods have also been explored for radio resource management, multi-access optimisation, traffic prediction, mobility management, and service quality assur-

ance [2]. These advances indicate a clear trend towards AI-native network intelligence, where data-driven models support monitoring, optimisation, and closed-loop control.

However, the effectiveness of ML models strongly depends on the availability of representative and high-quality datasets. In practical telecommunication networks, data collection and sharing remain difficult because operational data may contain sensitive information, including network topology, device configurations, service conditions, customer traffic, and operator policies. Although several open-source datasets [3] [4] have

been released for optical or wireless network research, most of them focus on a single domain, a specific task, or a limited experimental scenario. This restricts their applicability to future intelligent networks, where decisions increasingly rely on joint observations across multiple domains. The lack of open-source cross-domain datasets is particularly important for converged optical and wireless networks, where wireless access and optical transport domains are tightly coupled to support end-to-end services. Service degradation may arise from wireless channel variations, optical transmission impairments, traffic dynamics, or their combined effects. Therefore, cross-domain datasets are essential for end-to-end service monitoring, fault diagnosis, resource coordination, and AI-assisted orchestration. However, constructing such datasets is challenging due to heterogeneous infrastructures, telemetry interfaces, data formats, time scales, and control mechanisms. Privacy concerns in wireless telemetry and confidentiality issues in optical-domain topology and configuration data further complicate data collection, fusion, and sharing. Synthetic data generation can complement practical measurements by covering diverse network scenarios and rare operating conditions. Digital twins (DTs) provide an effective approach for generating synthetic network data by emulating network behaviour under configurable topologies, traffic patterns, device parameters, and service requirements. However, existing DT-based data generation still relies heavily on manual scenario design, parameter configuration, experiment execution, result analysis, and simulation refinement. This process is time-consuming and difficult to scale, especially for cross-domain networks with large configuration spaces and domain-specific constraints.

To address these challenges, this paper presents an open-source cross-domain dataset framework that supports both practical and generated datasets: practical data are organised through network telemetry and semantic abstraction, whereas synthetic data are demonstrated through an LLM-driven multi-agent DT workflow. The term cross-domain has two precise meanings in this work. First, independently acquired optical-transport, fibre-sensing, and wireless-access datasets are exposed through a common repository and metadata structure. Second, a constructed interoperability benchmark combines optical and wireless feature sequences to test whether compact wireless representations remain useful to a common downstream model. Because the practical campaigns were conducted at different times, this benchmark does not represent simultaneous end-to-end measurements and is not used to infer a physical causal relationship between the domains.

The main contributions are as follows. (1) We consolidate practical optical-transport, fibre-sensing, and multi-access wireless datasets, together with their acquisition context, formats, sampling information, and repository terms. The telemetry platform builds on our previous implementations [5, 6]; the present contribution is their dataset-level organisation and cross-domain interface. (2) We specify and evaluate hand-crafted and learned wireless semantic representations using chronological train/validation/test partitions, train-only preprocessing, repeated runs, raw/PCA/random baselines, and multiple privacy attackers. (3) We demonstrate an LLM-driven multi-agent interface to GNPY and ns-3 DT engines, compare the optical configuration set with a deterministic enumerator, and report an independently identified OpenAI reproduction with exact request/response traces and usage statistics. (4) We audit configuration uniqueness, result warnings, simulator provenance, and independent replayability using both a pinned historical-

compatibility replay and a current strict-schema check, thereby separating workflow feasibility, numerical reproducibility, and physical calibration.

The remaining sections are organised as follows: Related work on cross-domain data availability, network telemetry and semantic data abstraction are presented in Section 2, followed by an overview of DTs and LLM-Driven Network Automation. Section 3 presents the unified framework for cross-domain dataset collection and LLM-driven data generation. Section 4 details the proposed cross-domain telemetry architecture and practical datasets collected from optical and wireless testbeds. Then, Section 5 introduces the LLM-driven multi-agent DT framework and demonstrates its use for automated data generation. Finally, Section 6 concludes with a summary of this work and future research directions.

2. RELATED WORK

A. Cross-Domain Data Availability and Network Telemetry

With the increasing use of ML in telecommunication networks, data have become a key enabling resource for model training, validation, inference, and updating. In cross-domain optical-wireless networks, practical telemetry is particularly valuable because it captures communication performance, device states, service requests, and environmental variations under real operating conditions. In optical transport and fibre-sensing systems, such data may include bit error rate (BER), optical signal-to-noise ratio (OSNR), Q-factor, optical power, route information, configuration records, device alarms, state-of-polarisation (SOP) evolution, backscatter traces, distributed acoustic sensing signals, and vibration-related features [7–9]. In wireless access networks, standardised telemetry may include throughput, packet loss, delay, signal strength, cell load, mobility, traffic distribution, and per-device access statistics [10, 11]. The value of these heterogeneous data lies not only in individual measurements, but also in their temporal alignment and shared service context across optical transport, fibre sensing, and wireless access domains.

However, practical cross-domain data are difficult to collect, align, and share. Optical transport, fibre sensing, and wireless access systems rely on different infrastructures, telemetry interfaces, data formats, and time scales. In addition, operational data may expose real network topology, configuration, device states, customer traffic, service conditions, and operator policies, thereby limiting public data sharing. Consequently, only a limited number of open optical-network datasets are available. Representative examples include the Microsoft wide-area optical backbone dataset for operational network-performance analysis [4, 12], the Fraunhofer HHI QoT dataset collection for ML-assisted QoT estimation in elastic optical networks [13], and the Alibaba disaggregated production-network dataset for field-data-based QoT estimation and parameter optimisation [3, 14]. Recent dataset-oriented efforts have further covered space-division-multiplexing networks, fibre sensing, optical-device modelling, and radio-over-fibre systems [15–18]. Nevertheless, most existing datasets remain limited to a single domain, topology, task, or measurement campaign, and rarely provide time-aligned optical-transport, fibre-sensing, and wireless-access observations under a common service context.

Network telemetry provides the basis for scalable data collection [19]. Model-driven and streaming telemetry solutions based on NETCONF/YANG, gRPC, gNMI, and publish-subscribe mechanisms have been developed to expose network states and

performance measurements [19, 20]. In optical networks, prior studies have demonstrated telemetry services for real-time monitoring in disaggregated optical networks, streaming-telemetry-assisted diagnosis, Kafka-based optical monitoring frameworks, and unified monitoring platforms integrating telemetry collection, time-series storage, configuration databases, and streaming pipelines [5, 21–23]. In wireless networks, O-RAN and AI-native RAN architectures also rely on multi-source telemetry to expose radio, traffic, and service context to intelligent controllers and AI/ML applications [24]. However, most existing telemetry frameworks are designed for specific domains, vendor interfaces, or monitoring tasks. Cross-domain data collection, time alignment, format consistency, repository management, and service-level interpretation across optical transport, fibre sensing, and wireless access domains remain insufficiently addressed.

B. Semantic Data Abstraction for Privacy-Aware Cross-Domain Fusion

Cross-domain dataset generation requires not only collecting heterogeneous telemetry, but also deciding which information should be shared, stored, and fused. Directly aggregating raw wireless telemetry into a central repository may introduce unnecessary communication and storage overhead while exposing device-specific or service-sensitive information. This issue is particularly critical in multi-access wireless networks, where raw telemetry may reveal device identity, access behaviour, traffic patterns, mobility states, and service conditions [25]. Therefore, practical optical–wireless data fusion requires compact and task-relevant representations that preserve useful information for downstream analysis while reducing the exposure of sensitive raw observations.

Task-oriented semantic communication provides a useful principle for such abstraction by transmitting information relevant to downstream tasks rather than reconstructing all source observations with equal fidelity [26, 27]. Within telemetry systems, semantic representations can be constructed from expert-designed indicators or learned encoders. In particular, information-bottleneck methods can learn compact latent variables that preserve task-relevant information while discarding irrelevant input details [28]. Such representations can summarise wireless load, link stability, access diversity, congestion risk, and service degradation, thereby reducing data volume and enabling more efficient fusion with optical transmission and fibre-sensing measurements. However, compact representations are not automatically privacy preserving, because sensitive attributes may still be inferred from the transmitted features. Privacy-aware representation learning therefore requires information bottlenecks, noise injection, adversarial objectives, or representation regularisation to balance downstream utility against sensitive information leakage [28–31].

Existing studies on semantic-communication and privacy-aware representation mainly focus on single-domain communication data, end-to-end transmission efficiency, or downstream task accuracy. Limited work has investigated how wireless telemetry should be abstracted for cross-domain optical–wireless dataset generation, or how such representations should be evaluated jointly in terms of cross-domain utility, data reduction, and privacy exposure. This motivates the design of wireless semantic modules that transform raw multi-access telemetry into compact task-relevant representations for privacy-aware fusion with optical transmission and fibre-sensing data.

C. Digital Twins and LLM-Driven Data Generation

The preceding subsections concern the collection and selective sharing of practical telemetry. Synthetic data address a different limitation: even a well-instrumented testbed cannot economically cover every topology, loading condition, device setting, or rare failure. Synthetic data can therefore complement, rather than replace, independently collected measurements by expanding scenario coverage and providing labels under controllable conditions. Network DTs provide a natural execution tool because they emulate network behaviour under configurable topology, traffic, device, channel, and service conditions [32].

In optical networks, GNPpy provides an open and vendor-independent physical-layer modelling framework for emulating optical paths under configurable topology, equipment, channel loading, and transmission-parameter settings [33]. Recent studies have used GNPpy-based DTs to generate optical-network datasets under normal and degraded conditions, such as amplifier degradation, span-loss variation, and component misconfiguration [34, 35]. In wireless networks, simulators and DT platforms, such as ns-3 and Colosseum, support repeatable packet-level, radio frequency, and protocol-stack emulation [36, 37], while generative-AI-enabled wireless DTs have been explored for 6G modelling and scenario augmentation [38, 39]. These studies demonstrate the value of DTs for controllable data generation. Nevertheless, DT-based dataset generation still requires substantial manual effort to define objectives, configure scenarios, execute simulations, validate results, and refine parameter settings. This manual process becomes costly and difficult to scale in cross-domain optical–wireless environments, where different DT tools rely on different abstractions, interfaces, and configuration formats.

Recent studies have begun to integrate LLMs, AI agents, and DTs to reduce the manual burden of network automation. In optical-network DT research, modelling, telemetry, self-learning, and lifecycle management have been identified as key components for intelligent optical networks [40]. GPT-enabled DT assistants have also been demonstrated to interpret user intentions and interact with DT functions for autonomous optical-network management [41]. In addition, DTs have been used to provide updated network states and pre-verify LLM-generated strategies before deployment, with demonstrations covering performance optimisation, protection switching, and recovery after fibre cuts [42]. Other studies have explored fine-tuned local agents, DT-aided wavelength-loading control, AI-agent integration, and field-trial demonstrations for optical-network lifecycle automation [43–46]. These studies demonstrate the potential of LLMs and AI agents to interact with DTs, but their focus remains on network operation and management rather than autonomous dataset generation.

The cited DT assistants and AI-agent studies automate parts of modelling and network operation, but they still rely on explicitly supplied objectives and configuration ranges [41–44]. This leaves dataset generation as a distinct workflow-design problem. The present work therefore studies an LLM-driven interface that translates a high-level request into structured tasks, invokes optical and wireless DT engines, analyses generated KPIs, and proposes refinements. Section 5 evaluates this as a feasibility demonstration and includes a deterministic configuration-enumeration baseline; it does not claim that an LLM is necessary for exhaustive enumeration.

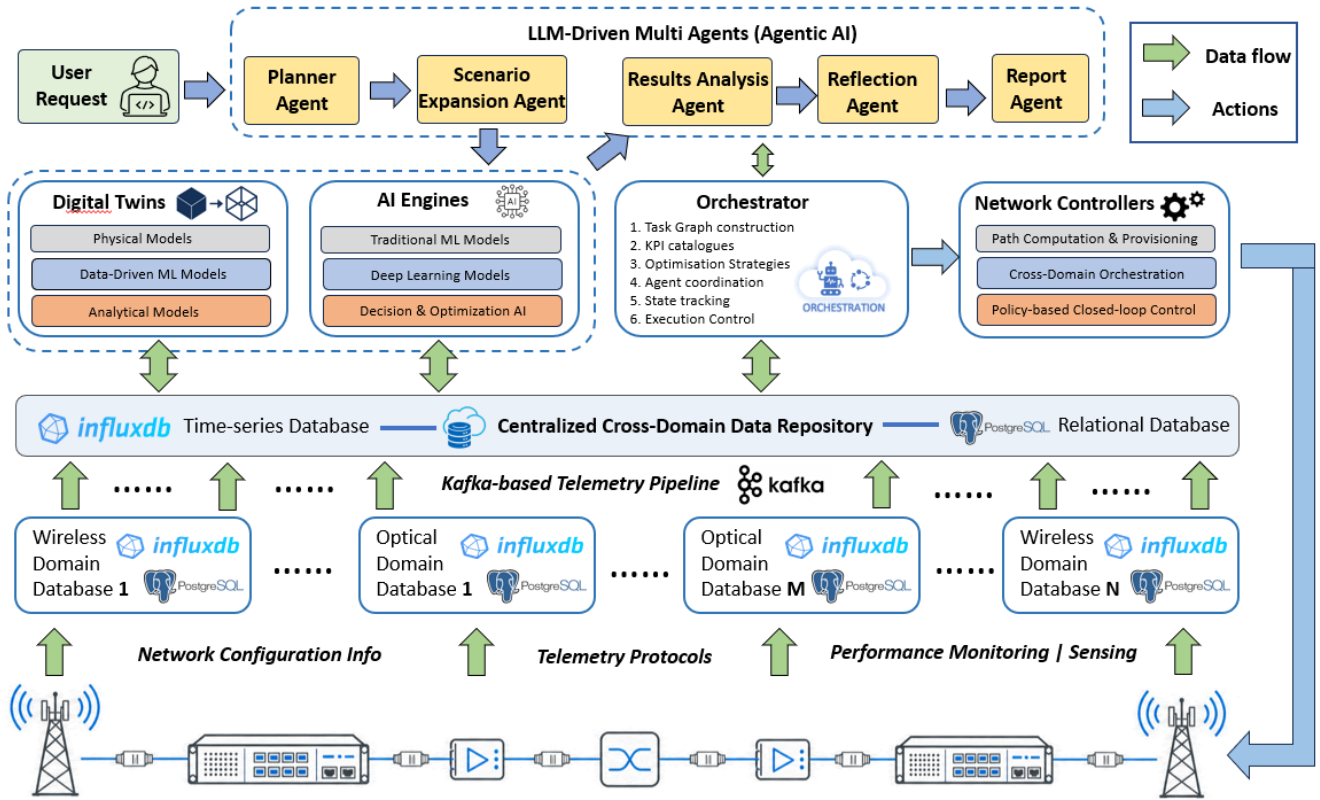


Fig. 1. Scope map of the complete dataset framework. The left branch organises independently acquired practical datasets; the right branch places the demonstrated DT data-generation workflow within a broader prospective network-intelligence context.

3. UNIFIED FRAMEWORK FOR CROSS-DOMAIN DATASET COLLECTION AND LLM-DRIVEN DATA GENERATION

Figure 1 illustrates a unified framework for cross-domain dataset collection and LLM-driven data generation in wireless–optical network environments based on our previous work [5] [6]. The bottom layer represents the physical infrastructure of a general end-to-end network scenario, in which wireless access domains are interconnected through an optical transport network. This abstraction captures a practical multi-domain networking environment where heterogeneous devices, protocols, and control entities coexist across wireless and optical segments.

The framework has two complementary branches. The practical-data branch collects and publishes independently acquired telemetry, while the generated-data branch converts a user request into simulator inputs and retained artefacts. Component-level storage and telemetry details are deferred to Section 4.

In this paper, the implemented path comprises telemetry export, semantic evaluation, orchestrated task construction, and GNP/ns-3 execution. AI engines and network controllers remain contextual integration points; QoT prediction, path optimisation, and physical-network enforcement by those blocks are not evaluated.

Figure 1 is therefore a scope map, whereas Fig. 6 later gives the evaluated data-generation sequence. The orchestrator dispatches that sequence but is not claimed to control every component in Fig. 1.

The detailed design of the cross-domain telemetry architecture across optical and wireless networks is presented in Section 4, while the LLM-driven multi-agent DT framework for

cross-domain data generation is further elaborated in Section 5.

4. CROSS-DOMAIN TELEMETRY ARCHITECTURE ACROSS OPTICAL AND WIRELESS NETWORKS

A. Proposed Cross-Domain Telemetry Architecture

Figure 2 presents the cross-domain telemetry architecture developed from our earlier telemetry platform [5]. The implemented elements used for the released practical datasets are device-data acquisition, time-series/configuration storage, and the Kafka transfer path. The semantic-module experiments use exported records from this platform. DT execution is demonstrated in Section 5; the AI-engine and controller blocks are architectural integration points and are not experimentally evaluated here.

A.1. Optical Transport Network Physical Layer

At the bottom of Fig. 2, the physical layer of the optical transport network consists of multiple interconnected optical nodes, including aggregation and intermediate nodes. These nodes represent practical optical transport infrastructure, in which NEs such as transponders, fibre links, optical switches, reconfigurable optical add-drop multiplexers (ROADMs), and optical amplifiers generate heterogeneous operational data.

The collected optical domain data can be categorised into three types: performance monitoring metrics, equipment parameters and topology information, and fibre sensing data. Performance monitoring metrics include receiver-side BER, OSNR, received optical power, amplifier input/output power levels, and per-port optical power measurements from optical switches. Equipment parameters and topology information include transmitter-side launch power, wavelength allocation, modulation format, amplifier gain, node configuration, and link

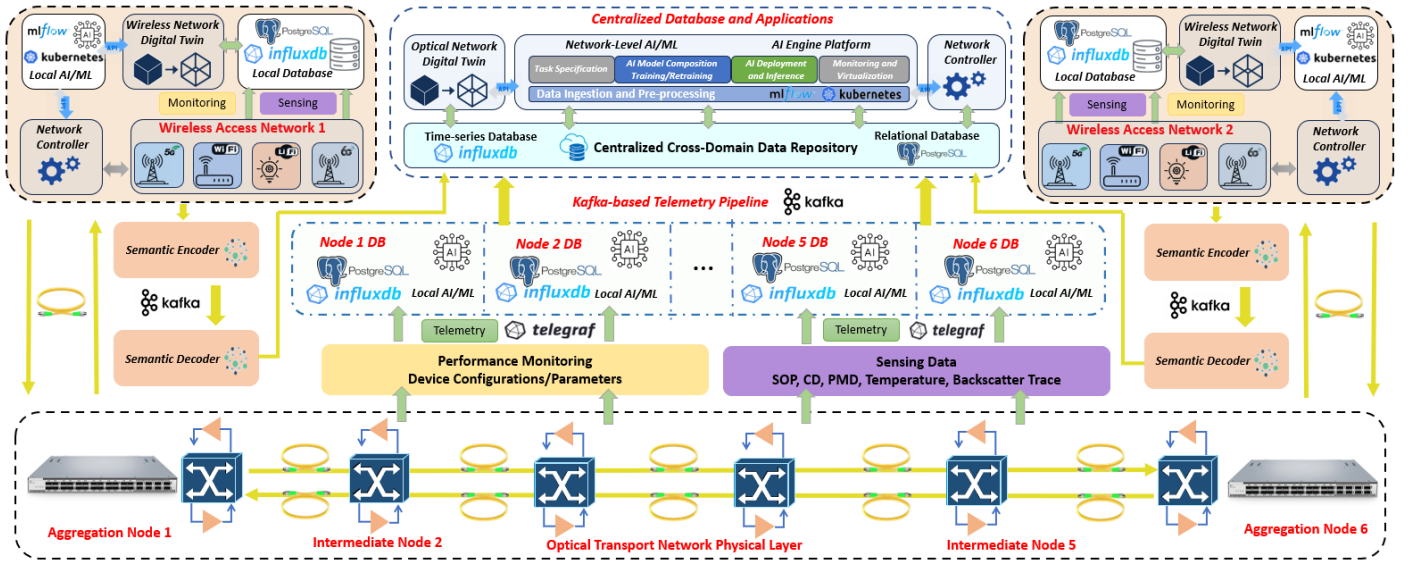


Fig. 2. Telemetry and repository view of the framework. Solid data paths correspond to implemented collection/storage functions; DT, AI-engine, and controller blocks indicate integration points, not evaluated closed-loop functions in this paper.

information. Fibre sensing data include SOP, chromatic dispersion (CD), polarisation mode dispersion (PMD), and backscatter traces. These data provide communication- and physical-layer visibility, supporting applications such as QoT estimation, anomaly detection, environment-aware operation, and proactive maintenance.

A.2. Distributed Node-Level Databases

Local databases store node- or domain-specific telemetry. In the implemented platform, continuously sampled BER, OSNR, optical-power, and sensing streams are stored in InfluxDB, whereas topology and equipment configurations are stored in PostgreSQL. Forecasting and anomaly-detection modules are possible consumers, but their deployment and scaling are outside the evaluation in this paper.

A.3. Telemetry Service Design

The implemented telemetry service ingests available NET-CONF, gRPC/gNMI, REST/HTTP, and vendor-interface records through Telegraf or source-specific collectors and writes timestamped values to the local databases.

For repository transfer, selected local records are published to Kafka topics and consumed by the central repository. Campaign and source identifiers are retained during aggregation so that independently acquired records are not mistaken for naturally aligned observations.

Kafka is a standard implementation component rather than a contribution. Because no throughput or latency benchmark is reported, this deployment is not used as evidence of scalability.

Before wireless data are shared with the central repository, the evaluated semantic modules transform raw telemetry into compact task-related representations. The experiment measures payload dimension, downstream utility, and attribute-inference risk; it does not provide a cryptographic privacy guarantee. The design and evaluation are detailed in Section B.

A.4. Centralised Cross-Domain Data Repository

The central repository provides a common access point for selected monitoring streams, topology, equipment, and service metadata. Importantly, a common repository does not imply

that all released campaigns are time aligned: timestamps and campaign identifiers are retained so users can distinguish contemporaneous records from independently acquired datasets.

A.5. Network Intelligence and Controllers

Above the repository, DTs provide the execution environments used in Section 5. AI engines and network controllers are included to show prospective consumers of the practical and generated datasets. Model lifecycle management, path provisioning, device reconfiguration, and closed-loop enforcement are not contributions evaluated by the present experiments.

A.6. Wireless Access Network Domains

The wireless-domain blocks in Fig. 2 show where 5G, Wi-Fi, and LiFi telemetry enters the common repository. In the present study, the implemented/evaluated wireless functions are testbed acquisition and semantic abstraction. Wireless DT state synchronisation, AI inference, and controller actions are prospective and are not reported as experimental results.

B. Semantic Modules for Cross-Domain Data Fusion

The semantic module is evaluated as an interface for converting wireless telemetry into compact representations. We compare V1, a hand-crafted representation; V2, a learned eight-dimensional latent representation; and V3, a learned three-dimensional bottleneck with optional training-time noise. The term privacy-aware refers to an evaluated design objective—dimensionality and noise are varied and attribute leakage is measured—rather than a formal privacy guarantee.

B.1. Semantic Module Designs

The input contains 12 features derived from 30-s wireless records: aggregate 5G, Wi-Fi, and LiFi traffic; total traffic; mean Wi-Fi and LiFi signal levels; mean Wi-Fi inactivity; a five-record rolling traffic fluctuation; three RAT-activity indicators; and a combined signal-quality score. RAT activity is defined relative to the 25th percentile of training traffic. All min-max and standard scalars are fitted on the training period only. Four task-oriented targets are then formed: load L , access diversity D , link stability S , and

congestion risk C . With q denoting combined signal quality, f normalised traffic fluctuation, and i normalised inactivity,

$$D = (a_{5G} + a_{WiFi} + a_{LiFi})/3, \quad (1)$$

$$S = [0.5q + 0.3(1 - f) + 0.2(1 - i)]_{[0,1]}, \quad (2)$$

$$C = \sigma(2L + 1.5i + f - 1.5S - 0.8D), \quad (3)$$

where $[\cdot]_{[0,1]}$ denotes clipping and $\sigma(\cdot)$ is the logistic function. These variables were selected to cover traffic intensity, temporal stability, multi-access availability, and a composite risk indicator using fields present across the wireless testbed exports. A confidence value $1 - f$ accompanies each transmitted representation. Fig. 3 summarises the three designs.

V1 transmits eight deterministic indicators: L , S , D , C , rolling-mean traffic, traffic fluctuation, signal quality, and inactivity risk. V2 uses a 12–64–32–8 encoder and a symmetric decoder to reconstruct L , S , D , and C . V3 uses a 12–32–16–3 encoder and a symmetric decoder. Its primary setting injects zero-mean Gaussian noise with $\sigma = 0.05$ at the encoder input during training. The encoder–decoders use mini-batch Adam, batch size 32, at most 200 epochs, and early stopping with patience 12 on validation loss. For downstream utility, the four decoded targets plus confidence are fused with optical features; the privacy attacker receives the latent code plus confidence. The source, exact seeds, preprocessing parameters, and per-seed outputs are released with the experiment scripts.

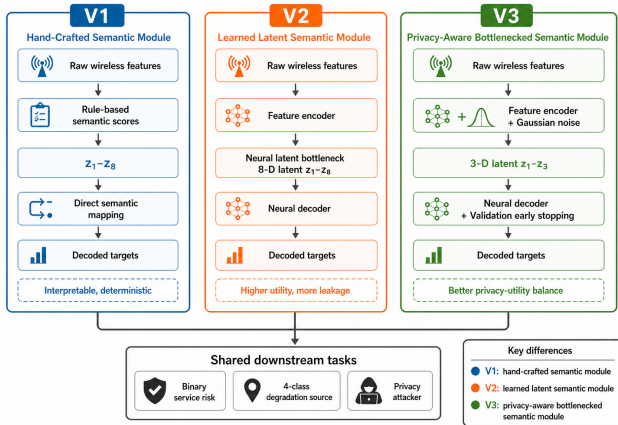


Fig. 3. Structures of three semantic-module designs.

The optical and wireless campaigns do not overlap in time. We therefore construct an explicitly labelled interoperability benchmark by mapping the ordered 20,960 optical records to the ordered 2,849 wireless records at uniformly spaced relative positions; no timestamp equality is asserted. This benchmark tests schema compatibility and representation utility, not contemporaneous end-to-end behaviour. After one-step target shifting, 2,848 samples remain. Optical risk is defined using the training-period 15th percentile of Q-factor or 85th percentile of BER. Wireless risk is defined as $C > 0.65$, $S < 0.35$, or $D < 0.34$. Binary service risk is their logical OR; four degradation classes denote neither, wireless only, optical only, and both. These are rule-derived benchmark labels rather than field-observed service failures.

B.2. Testing Results

Samples are partitioned chronologically into 60% training, 20% validation, and 20% test segments, with a ten-record purge gap before each boundary to prevent five-record rolling features

crossing partitions. This yields 1,698/560/570 evaluated samples. Every threshold, scaler, encoder, and downstream classifier is fitted after the split using training data only. Utility is measured with a 300-tree class-balanced random forest over ten seeds (3, 7, 11, 19, 23, 31, 42, 53, 71, and 97). We report Accuracy, Balanced Accuracy, Macro-F1, AUROC, and AUPRC; Table 1 shows the primary Macro-F1 results and 95% confidence intervals.

Privacy is operationalised as attribute inference from the transmitted representation. Dominant CPE and dominant RAT are defined by the largest aggregate traffic contribution in each record. Four attackers are evaluated: class-balanced logistic regression, a 300-tree random forest, a two-layer MLP (32 and 16 units), and histogram gradient boosting. Raw telemetry, PCA, random projection, majority, and stratified-random baselines are included. Because the test set contains 559 Wi-Fi and only 11 LiFi records (and no 5G record), RAT accuracy is dominated by the 0.981 majority baseline and is retained only as a diagnostic. CPE Macro-F1 is the primary privacy metric.

V3 reaches binary Macro-F1 0.6457 ± 0.0054 , similar to V1 (0.6455 ± 0.0040) and above raw wireless plus optical features (0.6329 ± 0.0044). The overlapping intervals do not support a claim that V3 is superior to V1. For the four-class task, raw features perform best. Moreover, the test distribution is 436/115/13/6, so the four-class result is exploratory. Removing V3 training noise slightly increases binary Macro-F1 from 0.6457 to 0.6473; increasing noise to $\sigma = 0.10$ lowers the strongest CPE-attack Macro-F1 to 0.2319. Thus, noise reduces observed leakage but does not improve utility in this dataset.

V3 therefore offers a useful operating point: it retains V1-level binary utility with moderately lower observed CPE leakage. It is not universally optimal—PCA3 leaks less but loses more utility, and raw features perform best on the imbalanced four-class task. We consequently describe V3 as privacy-aware and report the complete trade-off rather than claiming privacy preservation or statistical superiority.

C. Practical Dataset Description

Under the telemetry architecture, we release datasets from separate optical-transport, fibre-sensing, and wireless-access campaigns [47]. Table 2 summarises their scope and machine-readable files. The campaigns have different dates, sampling rates, and service contexts; “cross-domain” therefore refers to common publication and representation rather than native sample-level alignment. The practical-data repository terms permit academic, non-commercial research, but no standard licence file or SPDX identifier is supplied; the W4C repository likewise does not specify formal reuse terms. A Croissant 1.0-compatible JSON-LD record accompanies the revision artefacts and documents both 339-km Voyager campaigns, the processed wireless and constructed fusion tables, semantic utility and privacy results, the GNPpy audit and replay results, and the final W4C synthetic wireless release, including public locations, byte sizes, checksums, and field units where they are defined by the source files. The final W4C bundle is distributed under W4C_THREE_RAT_SWEEP/ in the revision repository [48].

C.1. One-Month Performance Monitoring Data from Optical Transport Network over NDFF

Figure 5(a) presents the experimental setup over part of the UK National Dark Fibre Facility (NDFF), including transponders, erbium-doped fibre amplifiers, wavelength-selective switches, and optical switches. Seven equalised 200-GHz-spaced PM-

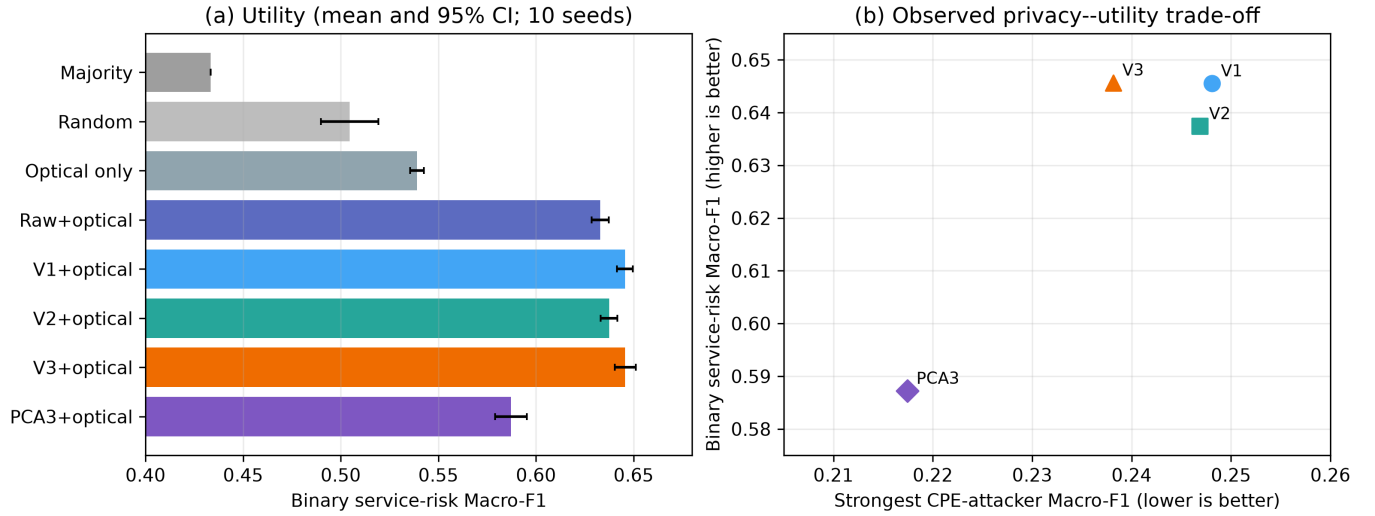


Fig. 4. Leakage-free temporal evaluation. (a) Binary service-risk utility; error bars are 95% confidence intervals over ten seeds. (b) Utility versus the strongest observed CPE attacker among four attacker families. Lower attacker Macro-F1 indicates less observed leakage but is not a formal privacy guarantee.

Table 1. Temporal-holdout utility and CPE-attribute leakage (mean $\pm$ 95% confidence interval over ten seeds).

Representation	Binary service-risk Macro-F1	Four-class Macro-F1	Strongest CPE-attack Macro-F1↓
Majority baseline	0.4334 $\pm$ 0	0.2167 $\pm$ 0	0.1235 $\pm$ 0
Stratified-random baseline	0.5046 $\pm$ 0.0147	0.2427 $\pm$ 0.0101	0.2530 $\pm$ 0.0095
Raw wireless + optical	0.6329 $\pm$ 0.0044	0.4161 $\pm$ 0.0051	0.2757 $\pm$ 0
V1 + optical	0.6455 $\pm$ 0.0040	0.3823 $\pm$ 0.0033	0.2481 $\pm$ 0
V2 + optical	0.6374 $\pm$ 0.0044	0.3958 $\pm$ 0.0175	0.2469 $\pm$ 0.0102
V3 + optical ($\sigma = 0.05$)	0.6457 $\pm$ 0.0054	0.3955 $\pm$ 0.0185	0.2382 $\pm$ 0.0113
PCA3 + optical	0.5872 $\pm$ 0.0081	0.3829 $\pm$ 0.0045	0.2174 $\pm$ 0.0059

16QAM/QPSK signals were generated by Facebook Voyager, ADVA FSP 3000 TeraFlex, and Ciena transponders. Voyager and ADVA support 16QAM and QPSK, whereas the Ciena device supports QPSK. The signals traversed the 339-km loop from Bristol through Bradley Stoke, Froxfield, Reading, and Power Gate, with carrier frequencies from 193.25 to 194.45 THz. The launch power was fixed at -5 dBm as the operating point used for this campaign; no optimality claim is made. A cloud configuration and monitoring database stored the field-trial measurements.

The public repository distinguishes two measurement campaigns on this same 339-km loop. The earlier release contains 1,373 processed records collected from 26th December 2023 to 26th January 2024. Its 10-s raw BER measurements were resampled to 15-minute intervals, and four Voyager-channel Q-factors were calculated using $Q = \sqrt{2} \operatorname{erfcinv}(2 \operatorname{BER})$. This preprocessing reduces short-term measurement noise and aligns the sampling interval with that of the Microsoft open-source dataset.

A separate Christmas 2025 campaign, used as the optical source for the constructed semantic-fusion benchmark in Section IV-B, contains 20,960 valid eight-channel BER records from 22nd December 2025 16:38:43 to 24th January 2026 16:10:55. These are real acquisition timestamps retained without shifting or anonymisation. The raw CSV preserves 38 embedded headers from 39 concatenated acquisition segments; the released preprocessing removes those header rows before timestamp and BER parsing. Consecutive valid records are mostly 70–72 s apart, with larger acquisition gaps, and this file was not resampled to 15-minute intervals. Its unchanged source file and SHA-256 checksum are published with the acquisition note [47]. Because

the two campaigns differ in date, channel count, and sampling, they are reported as separate products rather than as stages of one transformation.

We also release a complementary dataset from a previous 986-km NDFF field trial [49] comprising two fibre links and ROADM nodes, where multi-channel Q-factor measurements were collected for end-to-end QoT prediction. Together, the 339-km and 986-km datasets provide representative short- and long-link field-trial scenarios for QoT estimation, short-term performance forecasting and anomaly detection.

C.2. Four-Week Sensing Data from BDFI and M Shed

Figure 5 (b) illustrates the experimental setup for long-term fibre sensing based on bidirectional transmission. The experiment was conducted over two deployed urban fibre links in Bristol: a 7.776-km link connecting our lab in Merchant Ventures Building (MVB) to Bristol Digital Futures Institute (BDFI), and an 8-km link connecting MVB to M Shed Museum. Along the MVB-BDFI route, the fibre passes beneath multiple railway tracks near Bristol Temple Meads Station at the 3.2-km point, where it laterally traverses beneath a railway bridge. In contrast, the MVB-M Shed route does not pass beneath railway tracks.

An ADVA FSP 3000 Teraflex transponder generated two 32-Gbaud PM-QPSK optical signals, which were launched into the fibre links through optical circulators to enable counter-propagating transmission. The endpoint Circulators were interconnected through the live field fibre, with the forward and backward channels centred at 193.4 THz and 191.3 THz respectively. At each endpoint, the received optical signal was split using an optical coupler: 90% of the optical power was directed to

Table 2. Summary of the practical, constructed, and synthetic evaluation datasets used in this paper. Sizes refer to the principal processed files used in the revision experiments; raw sensing traces are distributed as multiple files.

Dataset	Records/sampling	Principal fields or labels	Format/processed size	Repository terms
339-km NDFP monitoring (2023–24)	1,373 processed records from 26 Dec. 2023 to 26 Jan. 2024; 10-s raw acquisition resampled to 15 min	Four Voyager BER channels and four derived Q-factors with timestamps	CSV and scripts; 0.15 MB	Academic research; non-commercial
339-km NDFP Voyager (2025–26)	20,960 valid raw records from 22 Dec. 2025 to 24 Jan. 2026; mostly 70–72-s spacing with acquisition gaps	Eight channel BER ratios and real timestamps; Q-factor derived in E1/E2	Raw CSV; 1.79 MB	Academic research; non-commercial
986-km NDFP QoT	Two links and ROADM nodes; record count and time coverage not available in the current release	12 channel states and Q-factors at three nodes; units follow the cited source	External CSV source cited in YangOFC2023; file size not available here	External reference; not in current release
Urban fibre sensing	Four-week campaign over 7.776-km and 8-km links; exact record count not available in the current release; polarimeter sampling at 97.656 kSa/s	BER, Q-factor, OSNR, received power, S_1 – S_3 , oscilloscope time/voltage; field units follow the source	MAT and CSV; file sizes not available here	External reference; not in current release
Wireless access	2,849 records over 24 h (nominal 30-s telemetry); 46 columns	CPE traffic, 5G temperature/traffic/power, Wi-Fi rate/inactivity/signal, LiFi rate/signal; raw units not declared in source	CSV; merged CSV 0.79 MB	Academic research; non-commercial
Synthetic GNP optical DT	8,192 executions; 6,144 unique configurations (6,120 valid and 24 no-signal); 96-case replay	Four paths, two modulation/spectrum settings, 256 binary occupancy patterns, three launch powers; GSNR, status, warnings, and replay differences	CSV, JSON, and JSONL; principal 6,144-row table 10.09 MB	Formal reuse terms not specified in current release; no SPDX identifier
Synthetic three-RAT wireless DT	48 configurations; 432 application-flow records (144 each for NR, Wi-Fi 802.11ax, and LiFi); 10-s runs	Per-flow KPIs (Mbps, ms, packets, bytes, ratios) and LiFi optical-state variables (m, rad, W, A, ratios); loads 2/5/10/20 Mbps, speeds 0.5/3 m/s, FOV 40°/70°, seeds 1–3	ns-3.44/5G-LENA; CSV, XML, JSON; public W4C bundle	Formal reuse terms not specified in current release; no SPDX identifier
Constructed fusion benchmark	2,848 one-step prediction samples; chronological split with 20 purged boundary records	Current/target source-row indices, deterministic wireless semantics, optical statistics, training-only thresholds, and rule-derived labels	CSV; 1.60 MB	Same repository terms

The final synthetic wireless fields and units are listed in the public W4C data dictionary. For source files not present in the current repository, record counts, file sizes, and field units are not inferred from citations or filenames.

the coherent receiver, while the remaining 10% passed through a polariser and was delivered to a polarimeter and a real-time oscilloscope equipped with a photodetector. This configuration enables simultaneous communication monitoring and fibre sensing. The polariser projects the received state of polarisation (RSOP) onto a fixed eigenstate, allowing environmental disturbances along the fibre to be captured through the measured polarisation dynamics. For vibration localisation, a common trigger from an arbitrary waveform generator was used to synchronise both the polarimeter and oscilloscope.

The dataset spans four weeks and is provided in MAT and CSV formats. It contains communication KPIs, including BER, Q-factor, and OSNR, and polarisation telemetry (Stokes parameters S_1 , S_2 , and S_3). These measurements were continuously collected by the ISIC end-to-end monitoring platform. Because the data originate from deployed urban fibres rather than laboratory spools, they capture environmental and operational variation. The dataset supports studies of SOP evolution, vibration detection and localisation, communication degradation under physical disturbance, ML-based fibre sensing, and communication-sensing integration.

C.3. 24-Hour Status and Performance Data from a Wireless Access Network Testbed

The wireless dataset was collected from a wireless access network testbed originally developed in the 5G-CLARITY project and further extended in the REASON project. These radio signals and telemetry records originate entirely within that wireless testbed; they are not converted from, transported as, or synchronised to signals on either optical field link. The optical interconnection shown in Figs. 1 and 2 is a system-level architecture, not the signal source for this campaign. As shown in Fig. 5(c), the testbed integrates 5G New Radio, Wi-Fi 6, and LiFi. Four fixed CPEs were deployed: CPE1–CPE3 were dual-homed to 5G and Wi-Fi, while CPE4 was tri-homed to 5G, Wi-Fi, and LiFi. All CPEs used MPTCP v1 in fullmesh mode with the de-

fault BLEST scheduler. During collection, CPE1–CPE3 generated randomised iPerf traffic towards CPE4.

The dataset provides 24 hours of time-aligned wireless telemetry, covering per-interface CPE traffic, 5G cell metrics, LiFi cell metrics, and Wi-Fi cell metrics. Specifically, it includes per-interface traffic distribution, 5G temperature, throughput and power, LiFi transmit/receive rates and signal power, and Wi-Fi bitrates, inactivity time and average signal levels. This dataset offers a comprehensive view of heterogeneous wireless performance and supports research on AI-driven multi-access optimisation, multi-connectivity control, MPTCP scheduling, and cross-technology coordination in 5G and beyond networks. By capturing the complementary characteristics of 5G, Wi-Fi, and LiFi in a unified testbed, it also provides a basis for wireless DTs, semantic telemetry abstraction, and cross-domain optimisation with optical datasets.

5. LLM-DRIVEN MULTI-AGENT DIGITAL TWINS FOR AUTOMATED CROSS-DOMAIN DATA GENERATION

While the preceding section presents practical cross-domain datasets collected from real optical and wireless testbeds, continuously acquiring sufficient real-world data for diverse or newly emerging network conditions remains challenging. To complement practical telemetry data and support timely ML model training and updating, this section introduces an LLM-driven multi-agent DT framework for automated cross-domain data generation.

A. Proposed LLM-Driven Multi-Agent Digital Twin Framework

Figure 6 illustrates the demonstrated data-generation workflow. Users specify the dataset objective, network scope, transmitter constraints, channel configurations, and modulation formats in natural language. The orchestrator passes a shared structured task state through planning, scenario expansion, DT execution, analysis, reflection, and reporting stages.

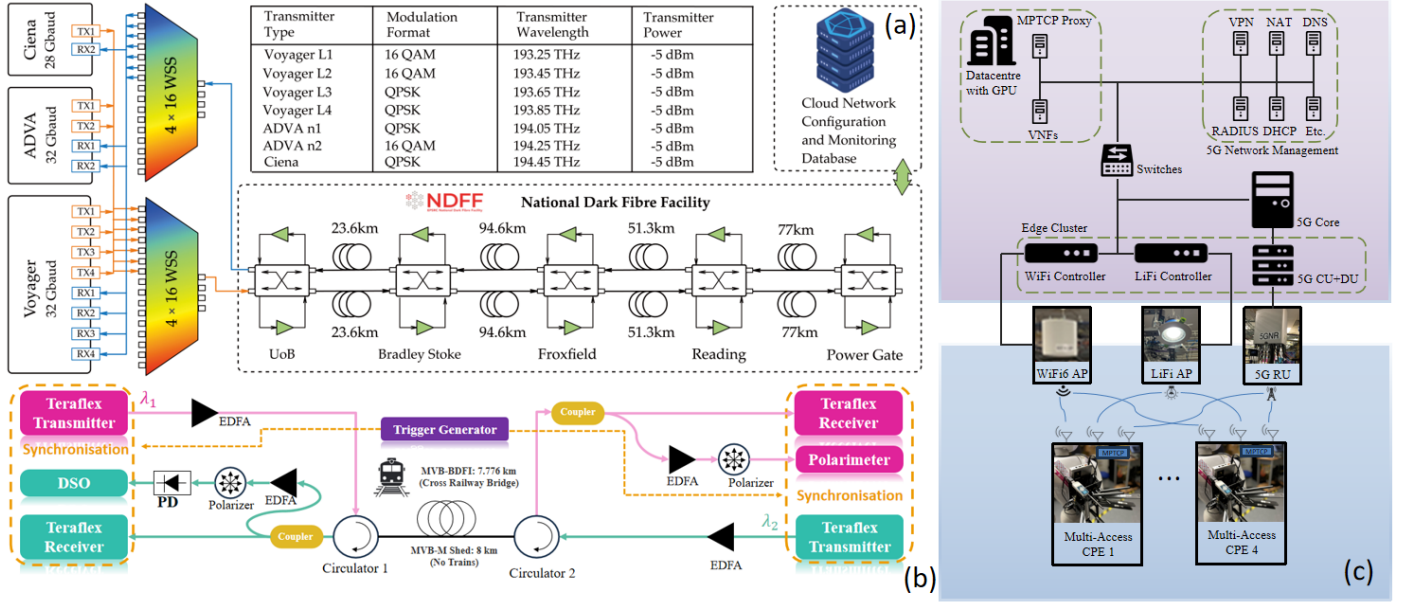


Fig. 5. Experimental setups for practical dataset collection: (a) optical transmission performance monitoring over the NDFF, (b) bidirectional fibre sensing over deployed urban fibre links, and (c) multi-access wireless network testbed.

Topology, device configurations, KPI definitions, constraints, and previous execution artefacts form the network knowledge base. The GNPpy demonstration reads JSON topology/equipment files and writes scenario-level request, result, and log artefacts. The physical network and database blocks in Fig. 6 provide context; they are not exercised as a live closed-control loop in this study.

The DT layer is an interface to execution engines. The reported experiments invoke the physics-based GNPpy model and the packet-level ns-3 simulator. Data-driven and decision/optimisation models shown in the architecture are prospective plug-ins and are not evaluated in this study.

The five named “agents” are functional roles within an orchestrated pipeline rather than independently persistent autonomous entities. The Planner produces the structured study specification; Scenario Expansion proposes executable cases; Results Analysis aggregates KPIs and logs; Reflection proposes changes to the search space; and Report formats the retained artefacts. Recovered source code shows that Planner, Scenario Expansion, and Reflection invoke an LLM, whereas Results Analysis and Report are deterministic modules. Simulator wrappers and the knowledge-base interface are their tools, while the orchestrator-owned task state is their shared memory. The roles were separated to give each transition an explicit input/output schema and to retain inspectable intermediate artefacts. This modularity is an engineering rationale, not evidence that each role requires an LLM; deterministic functions could replace them, and the released evidence does not quantify the communication overhead of role separation. The implementation does not expose separate long-term memories or independent subagent schedulers, so we do not claim those capabilities.

The orchestrator receives the user request, invokes all five roles in sequence, calls the DT wrapper, and records intermediate state. The arrows beneath the roles in Fig. 6 represent access to this shared orchestration layer, not direct control by only the final roles; its lower-right placement denotes a supporting control plane rather than the start or end of the execution order.

Reflection may trigger another scenario-expansion/execution pass. The experiments below establish that this workflow can produce artefacts for two DT engines; they do not establish lower cost, lower latency, or greater correctness than a conventional script.

B. Evaluation Protocol and Reproducibility Scope

We evaluate the corrected GNPpy demonstration against an independently implemented deterministic enumerator. The expected Cartesian product is specified independently of the observed records: four Bristol-origin destinations, Voyager transmitters, QPSK/37.5-GHz-slot and 16QAM/50-GHz-slot configurations, all 256 eight-slot occupancy patterns, and launch powers -6.5 , -6.0 , and -5.5 dBm, giving 6,144 unique configurations. Set precision and recall use source, destination, transmitter, modulation, occupancy pattern and power as keys; iteration and scenario identifiers are excluded. Repeated executions are counted separately, and the corrected second-round records supply the unique numerical dataset. The deterministic program performs configuration enumeration only; it does not replace natural-language intent parsing or GNPpy propagation.

The recovered orchestration snapshot supplies the Planner, Scenario Expansion, and Reflection prompt templates and JSON constraints. The authors report OpenAI as the historical provider; this attribution is distinct from verification by retained API records. The recovered OpenAI path specifies gpt-4o-mini, an optical-network-expert system message, temperature 0.2, a 90-s timeout and at most three attempts. However, the recovered directory has no Git metadata, and the historical archive does not retain a dated model snapshot, rendered per-call requests/responses, actual attempts, token use, cost, latency or human-intervention history. These historical measurements remain unavailable. The demonstration is not a repeated-trial comparison or an ablation of agent roles.

To provide prospective reproducibility without relabelling a new run as historical evidence, we performed a separately identified instrumented OpenAI reproduction on 13 September 2026.

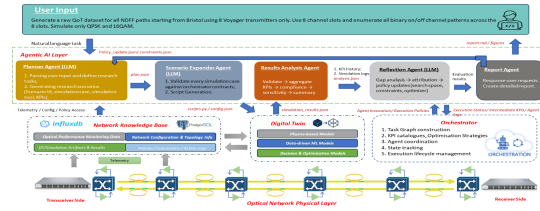


Fig. 6. Evaluated data-generation sequence and its context. Unlike the scope map in Fig. 1, this figure details the orchestrator-owned task state, role-specific LLM invocations, simulator interface, and retained artefacts.

It requested gpt-4o-mini-2024-07-18, temperature 0.2, a 90-s timeout, at most three attempts per call, and strict mode without silent rule-based fallback after an API failure. All six responses returned that model identifier. Planner, Scenario Expansion and Reflection each had two calls; all six API attempts succeeded on their first attempt. The retained requests/responses record 108,308 tokens (106,041 input and 2,267 output), 26.22 s summed API latency (2.88–7.61 s per call), and an estimated \$0.01727 text-token cost using the execution-date pricing snapshot. The second expansion request alone contains 96,716 input tokens. These are observations from one reproduction, not an estimate of general reliability or evidence of low communication overhead. The evidence archives are released at commit 82914127b70acf3b220bc09a56a179cb7abd1252 [50]; manifests retain the original dirty-source state, source/input hashes and subsequent repairs.

The workflow required manual intervention. An earlier invocation completed two API calls (4,057 tokens; estimated \$0.0009339) but was stopped after 344 GNPY import failures and one expected all-off result. Subsequent repairs addressed package installation, inactive-spectrum representation, CLI received-power extraction and a single-channel GNPY defect. The new two-round reproduction was also interrupted and resumed using manifest-hashed local topology/equipment files; recovery required a log-callback correction and rerunning nine failed single-channel cases. Its 2 h 1 min 25.3 s start-to-finish interval includes interruption, troubleshooting and resume; uninterrupted compute time and total human effort were not measured. The known API subtotal across the two invocations is \$0.01820025, excluding computation, labour and unrecorded activity. These costs are estimates rather than invoices. The six-call failure rate of 0/6 is an API-level observation; it does not imply an error-free or unattended workflow.

On 14 September, we corrected a deterministic post-processing defect that had overridden the second-round power sweep. The saved Reflection 1, Planner 2 and Scenario Expansion 2 responses consistently specified $[-6.5, -6.0, -5.5]$ dBm.

Table 3. Instrumented OpenAI reproduction with manual recovery (six-call invocation).

Item	Observed value
Model/settings	gpt-4o-mini-2024-07-18; temperature 0.2; timeout 90 s; maximum three attempts; strict mode
Role calls	Planner 2; Scenario Expansion 2; Reflection 2
API outcomes	6/6 successful first attempts; 0 retries; 0 API failures
Token usage	106,041 prompt; 2,267 completion; 108,308 total
API latency	26.22 s summed; 2.88–7.61 s per call
Estimated cost	\$0.01727 using the execution-date pricing snapshot
Retained trace	Six rendered requests, six raw responses, call log, manifest, source/input hashes, and archive SHA-256
Workflow interval	2 h 1 min 25.3 s, including interruption and recovery
Corrected replay	85 min 9.0 s; 0 new API calls; prior responses reused

The correction rejects inconsistent power sets and avoids inferring a sweep boundary from a single observed power. Replaying the saved suggestions regenerated 6,144 second-round records in 85 min 9.0 s, with no new LLM API calls; the 2,048 first-round records were reused. This replay is separate from the preceding run interval and API cost. The original Reflection 2 used pre-correction results and recommended another refinement, but the configured limit was two iterations. It was not recomputed on the corrected dataset, so neither convergence nor autonomous error repair is claimed.

C. Use Cases and Demonstration Results

To validate the feasibility of the proposed LLM-driven multi-agent DT framework, we implemented automated data-

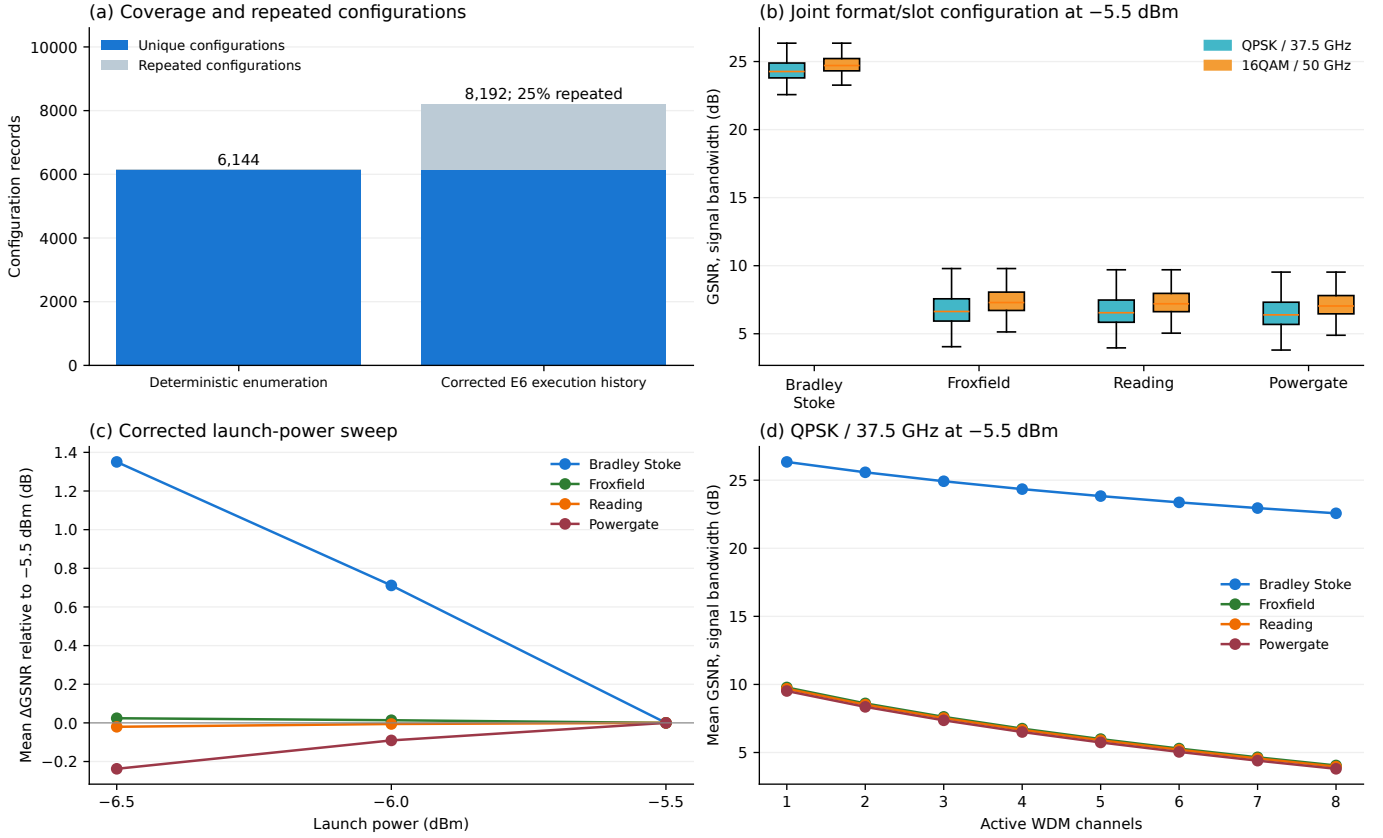


Fig. 7. Corrected GNPpy feasibility demonstration over the NDFF topology. (a) Configuration enumeration and execution-history counts; (b) QPSK/37.5-GHz and 16QAM/50-GHz GSNR distributions at -5.5 dBm; (c) route-dependent mean GSNR change over $[-6.5, -6.0, -5.5]$ dBm; and (d) QPSK mean GSNR versus active-channel count at -5.5 dBm. Numerical panels use the 6,120 ok configurations from the unique corrected second-round set; 24 all-off configurations are excluded from metric summaries. GSNR is measured in signal bandwidth. Each configuration first averages active-channel metrics, then configurations receive equal weight. Boxes show quartiles/medians with 1.5-IQR whiskers and omitted outlier markers; they are not repeated-run confidence intervals.

Table 4. Configuration-set baseline for the corrected GNPpy demonstration.

Method	Executions	Unique	Duplicate rate
Deterministic enumerator	0	6,144	0
Corrected E6 history	8,192	6,144	25%

Both sets have precision and recall 1.0 against the independently specified 6,144-key set. Configuration enumeration alone took a median 0.00145 s over 31 timing repeats on Python 3.12.10/Windows, with no LLM/API calls or simulations. These are timing repeats, not independent LLM trials. This duration excludes I/O, intent parsing and GNPpy and is not an end-to-end comparison.

generation workflows in both the optical transport and wireless access domains. Specifically, GNPpy and ns-3 were used as the DT execution engines for optical-domain and wireless-domain simulations, respectively. The following subsections present the corresponding data-generation use cases and demonstration results. The final synthetic wireless source and results are released in the cited revision repository under W4C_THREE_RAT_SWEEP/ [48]. The separate upstream framework repository's NS3_Wireless/ directory contains an earlier baseline and is not the source of the W4C results reported here.

C.1. GNPpy-Based Network-Level QoT Data Generation

In this use case, the workflow generates an optical QoT artefact based on the NDFF topology. The request covers paths from

Bristol, Voyager transmitters, eight WDM slots, and QPSK and 16QAM configurations. The Planner forms the study specification, and Scenario Expansion constructs the candidate paths, binary channel-occupancy patterns, and launch-power settings for GNPpy execution.

As illustrated in Fig. 7(a), four paths, two joint modulation/spectrum configurations and $2^8 = 256$ occupancy patterns produce 2,048 cases per launch power. Iteration 1 contains 2,048 records at -5.5 dBm. The corrected second round has 6,144 records at -6.5 , -6.0 and -5.5 dBm, repeating all first-round configuration keys. The combined 8,192 records comprise 8,160 ok and 32 expected no_signal outputs, but reduce to 6,144 unique configurations: 6,120 ok and 24 all-off configurations. The corrected set has zero missing or extra keys relative to the enumerator. All 6,120 active second-round scenarios agree in power across scenario metadata, request JSON, the CLI argument and returned metric metadata; all-off configurations do not invoke propagation.

Figure 7(b)–(d) uses the corrected second-round unique records. At -5.5 dBm, mean QPSK GSNR decreases from 26.34 to 22.57 dB on Bristol–Bradley Stoke and from 9.52 to 3.80 dB on Bristol–Powergate when active-channel count increases from one to eight. Over both joint format/slot configurations and all active occupancy patterns, Bradley Stoke mean GSNR decreases from 25.95 to 24.60 dB as launch power rises from -6.5 to -5.5 dBm, whereas Powergate increases from 6.65 to 6.89 dB.

Table 5. Historical-archive GNPY replay over 96 stratified cases.

Environment	Success	Mean $ \Delta\text{GSNR} $	Max $ \Delta\text{GSNR} $
v2.12 compatibility	96/96	0.00015 dB	0.00050 dB
v2.14.2 schema-fixed	96/96	0.628 dB	1.435 dB

These descriptive effects depend on route and retained equipment settings; the three-point sweep does not identify a globally optimal power.

The corrected second-round audit finds EDFA-below-minimum-gain warnings in all 6,120 CLI result files (48,960 occurrences) and unmet-ROADM-target-power warnings in all 6,120 files (12,240 occurrences). Thus, ok indicates returned numerical metrics, not warning-free or calibrated physical behaviour. These corrected-archive counts are distinct from the earlier historical audit.

Separately, we performed three checks on the historical published artefact. Its 8,160 result files contain EDFA minimum-gain warnings in every file (65,280 occurrences) and unmet ROADM target-power warnings in 6,120 files (12,240 occurrences). Its initial equipment file fails GNPY 2.13/2.14 YANG validation because `target_extended_gain=30` and `max_loss=0` are outside accepted ranges and duplicate transceiver-mode keys are present. After recovering legacy inputs, we independently replayed 96 cases stratified by path, joint format/slot configuration, power and channel load. Official GNPY v2.12 at commit 7ce665010970f57e46672db1ec0864ae448077e6 completed 96/96 and reproduced archived GSNR values with mean/max absolute differences of 0.00015/0.00050 dB. Because the recovered directory lacks Git metadata, this is an output-compatible reconstruction, not proof of the original commit. This historical numerical audit is not independent validation of the corrected E6 power sweep or its local single-channel patch.

For a current strict-schema check, we changed the copied span values to the GNPY reference defaults 2.5 and 28, removed unused duplicate 64-Gbaud teraflex modes, and represented disabled channels without the legacy -120 -dBm placeholders. GNPY 2.14.2 then completed 96/96 cases. Relative to the archive, its mean/max absolute GSNR errors were 0.628/1.435 dB and its mean/max OSNR errors were 0.209/0.929 dB. The mean GSNR difference was only 0.017 dB on the one-span Bradley Stoke route but approximately 0.83 dB on each longer route. Configuration warnings persisted in both replay arms; consequently, the first arm establishes numerical compatibility and the second exposes version sensitivity, but neither establishes warning-free physical calibration.

Finally, the corrected QPSK and 16QAM configurations use 37.5- and 50-GHz slots, respectively. Across 3,060 otherwise paired active configurations in the corrected set, the mean/max absolute signal-bandwidth GSNR differences are 0.550/1.088 dB. These are joint modulation/spectral-spacing effects, not a modulation-only comparison; the historical audit's 0.478/0.808 dB statistics apply to its different archived set.

Overall, the use case demonstrates conversion of a high-level request into structured simulator artefacts and execution of a reflection-proposed second-round sweep after deterministic software repair. It does not establish superiority over enumeration, unattended operation or Reflection convergence. The historical numerical replay and corrected configuration-consistency audit answer different reproducibility questions; neither removes the

need to calibrate equipment settings before making physical-accuracy claims. Larger topologies, independent repeated requests and architecture ablations remain outside this demonstration.

C.2. ns-3-Based Multi-Access Network Performance Data Generation

The LLM-driven workflow constructs and executes a bounded three-access synthetic wireless scenario at the same technology-class level as the practical multi-access testbed: 5G NR, Wi-Fi 6/IEEE 802.11ax, and LiFi. The generated dataset is separate from the measured wireless telemetry. It demonstrates configurable synthetic data generation and workflow interoperability, not calibrated emulation of the physical testbed and not validation of measured wireless values.

Each scenario contains three common mobile CPEs, each carrying one interface for each access branch. The NR branch uses CTTC 5G-LENA version 4.0.y on ns-3.44 with one gNB and three UEs, a 3.5-GHz carrier, 40-MHz bandwidth, numerology 1, 30-dBm gNB transmit power, and the 3GPP InH-OfficeOpen channel. The Wi-Fi branch uses native WIFI_STANDARD_80211ax with one AP and three stations, a 5-GHz 20-MHz channel, fixed 16-dBm transmit power, and an explicit constant-rate HeMcs0 configuration. CPE mobility is RandomWalk2d within 50×50 m bounds at 0.5 or 3 m/s. All branches use UDP downlink traffic with 1,024-byte packets, a common offered load of 2, 5, 10, or 20 Mbps per application flow, a 1.0–9.5 s application window, and a 10-s simulation stop.

The LiFi branch is an equation-driven LOS LiFi/OWC link model. It computes Lambertian emission, transmitter/receiver geometry, receiver FOV, received optical power, photodetector responsivity, shot and thermal noise, electrical SNR, OOK BER, and packet-error probability. A stock ns-3 point-to-point packet path is used as the IP/UDP access abstraction, while packet corruption is driven by the optical link calculation. This is not a complete standardised LiFi PHY/MAC and its explicit parameters are uncalibrated to the practical testbed.

For half-power semi-angle Φ_{half} , detector area A , distance d , irradiance angle ϕ , incidence angle ψ , filter gain T_s , and concentrator gain $g(\psi)$, the Lambertian order is $m = -\ln(2) / \ln(\cos \Phi_{\text{half}})$ and

$$H_{\text{LOS}} = \frac{(m+1)A}{2\pi d^2} \cos^m(\phi) T_s g(\psi) \cos(\psi), \quad (4)$$

when the irradiance geometry is valid and ψ is within the receiver FOV; otherwise $H_{\text{LOS}} = 0$. The concentrator is $g(\psi) = n^2 / \sin^2(\Psi_c)$ within the FOV, and received optical power is $P_r = P_t H_{\text{LOS}}$. The receiver current and the shot/thermal-noise terms determine SNR, which is mapped to $\text{BER} = Q(\sqrt{\text{SNR}}/2)$ and $\text{PER} = 1 - (1 - \text{BER})^L$. These equations follow the standard LOS optical-wireless link-budget formulation [51, 52] and are implemented in the source archived under W4C_THREE_RAT_SWEEP/ns3/ [48].

Four offered loads, two mobility speeds, two LiFi receiver FOVs (40° and 70°), and seeds 1, 2, and 3 yield $4 \times 2 \times 2 \times 3 = 48$ run-level configurations. The 40° setting was selected by a preflight over 30° , 40° , 50° , and 70° ; it produced measurable out-of-FOV states without near-total outage under the declared geometry. Each final configuration produces nine application flows (three branches $\times$ three CPEs), giving 432 application-flow records. The 624 raw FlowMonitor rows additionally include infrastructure/control flows, which are excluded from application statistics. Outputs comprise FlowMonitor XML, per-flow KPI

Table 6. Validation of the ns-3 synthetic wireless dataset.

Validation	Coverage	Result
Backend qualification	5G-LENA NR and native 802.11ax branches	Build/runtime PASS in the qualified ns-3.44 environment
LiFi equation tests	Independent Python reference; 9 tests	9/9 PASS
Compiled LiFi reference	3 geometries; 15 values	15/15 comparisons PASS
Integrated three-RAT mapping	1 final-source acceptance; 9 application flows	3 NR + 3 Wi-Fi + 3 LiFi; 9/9 mapped
Same-seed repeatability	1 final-source configuration executed twice	Byte-identical KPI, XML, mapping, metadata, and optical-state outputs
Full final sweep	48 configurations; 624 raw / 432 application flows	48/48 complete; 432/432 mappings pass

The rows distinguish backend qualification, independent equation/reference checks, integrated flow mapping, fixed-seed repeatability, and bounded-sweep completion under the declared scenario. None is physical-model validation.

CSV, scenario/configuration metadata, explicit flow mappings, sampled LiFi optical states, and aggregated CSV files.

KPI Definitions and Interpretation. For received bytes B_{rx} , transmitted/received packet counts N_{tx} and N_{rx} , and FlowMonitor time quantities in seconds, the exported throughput is $8B_{rx}/[10^6(t_{lastRx} - t_{firstTx})]$ Mbps. Mean delay is $10^3 \text{ delaySum}/N_{rx}$ ms; mean jitter is $10^3 \text{ jitterSum}/(N_{rx} - 1)$ ms for $N_{rx} > 1$. The FlowMonitor-reported packet-loss ratio is $\text{lostPackets}/N_{tx}$. Because traffic and simulation terminate at finite times, this field is not interpreted as terminal packet loss; in the final W4C application rows, 172 of 432 have $N_{tx} - N_{rx} \neq \text{lostPackets}$. No terminal-loss field is reported.

Validation of the Synthetic Wireless Dataset. Successful execution alone does not establish generated-data correctness. We therefore evaluated the final W4C stack at the backend, equation, integrated-flow, repeatability, and full-sweep levels (Table 6). The NR/5G-LENA and native 802.11ax branches passed the W4B qualification and W4C final-source build/runtime checks. The independent LiFi mathematical tests passed 9/9 checks, and compiled ns-3 link-probe values agreed with the independent Python reference in 15/15 comparisons across aligned-near, aligned-far, and out-of-FOV geometries. The integrated final-source acceptance run produced three flows per branch and 9/9 mapped application flows.

The final-source configuration was then executed twice with the same ns-3.44 build, seed, run number, workload, mobility, and FOV. The two outputs were byte-identical for the run configuration, scenario metadata, flow mapping, KPI CSV, FlowMonitor XML, and optical-state CSV. Finally, the bounded sweep completed all 48 configurations, with 432/432 application-flow mappings passing and no missing, extra, or duplicate configuration keys. These checks support correctness and reproducibility of the generated synthetic telemetry under the declared assumptions; they do not establish physical-testbed calibration or physical-model fidelity.

Under the adopted simulation configuration, increasing the offered load from 2 to 20 Mbps increased run-level throughput by 18.486 Mbps for NR and 15.726 Mbps for LiFi. For Wi-Fi 802.11ax, the corresponding throughput increase was only 0.151 Mbps, while mean delay increased by 1,653.88 ms and the FlowMonitor-reported packet-loss ratio increased by 0.701. These results indicate a configuration-specific queueing response in the Wi-Fi branch and load-responsive throughput in the NR and LiFi branches. Increasing mobility from 0.5 to 3 m/s

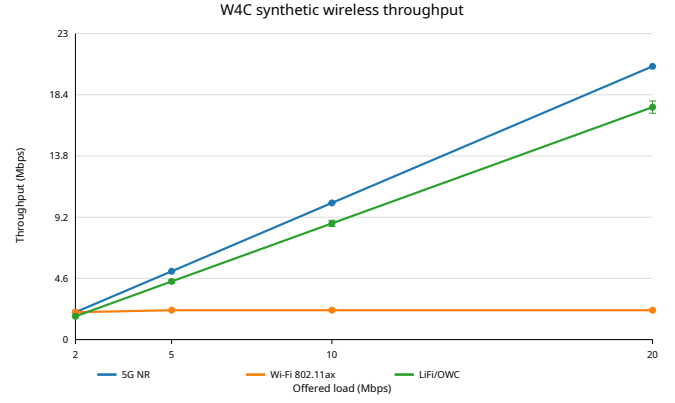


Fig. 8. Final W4C synthetic wireless throughput versus offered load. For each load and branch, the three CPE flows are averaged within each run, nuisance speed/FOV conditions are averaged within each seed, and error bars show the standard deviation across the three seeds.

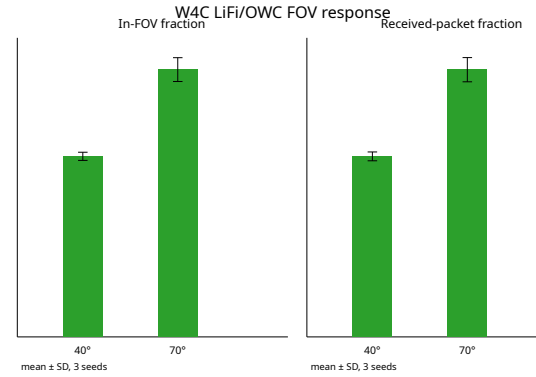


Fig. 9. Final W4C LiFi/OWC response to receiver FOV. The in-FOV fraction is computed from sampled optical states; the received-packet fraction is computed from exported LiFi flow counters. Bars show mean $\pm$ SD across three seeds after CPE and nuisance-condition averaging.

changed throughput by 0 Mbps for NR, 0.00018 Mbps for Wi-Fi 802.11ax, and -2.572 Mbps for LiFi under the selected geometry and channel models. Narrowing the LiFi FOV from 70° to 40° reduced the in-FOV fraction by 0.299 and the received-packet fraction by 0.299 in matched run-level comparisons. Because the ideal concentrator gain also changes with FOV, aggregate H_{LOS} and SNR are not asserted to decrease monotonically. Branch comparisons depend on topology, channel/helper assumptions, link rates, delays, offered load, mobility, and LiFi modelling choices; they are not universal technology rankings.

Wireless Scope and Limitations. The final synthetic backend uses 5G NR via CTTC 5G-LENA, Wi-Fi 802.11ax, and an equation-driven LOS LiFi/OWC model. The LiFi model is uncalibrated, uses a stock packet-access abstraction, and is not a complete standardised LiFi PHY/MAC. The NR, Wi-Fi, and LiFi comparisons are configuration-specific and do not imply universal technology superiority or calibrated reproduction of the practical testbed. The quantity $\text{lostPackets}/N_{tx}$ is reported only as the FlowMonitor-reported packet-loss ratio and is not interpreted as terminal packet loss at the finite simulation stopping time.

The validation supports implementation correctness and reproducibility of bounded synthetic telemetry, not physical-model fidelity or validation of the measured wireless dataset.

6. CONCLUSIONS

We presented a dataset framework containing independently acquired practical optical, sensing, and wireless campaigns and demonstrated DT-generated artefacts. Cross-domain in this work means common publication, metadata, representation, and constructed task-level fusion; it does not mean that the practical campaigns are simultaneous end-to-end measurements. On the constructed benchmark, the three-dimensional V3 representation preserves V1-level binary utility and moderately reduces observed CPE-attribute leakage relative to raw telemetry and V1. PCA leaks less but loses more utility, and V3 does not provide a formal privacy guarantee.

The LLM-driven workflow is retained as a feasibility demonstration with the existing Agent architecture. The corrected GNPY case contains 8,192 execution records and 6,144 unique configurations, exactly matching a deterministic enumerator. A separately identified new reproduction provides six complete request/response pairs, 108,308 tokens, 26.22 s summed API latency and an estimated \$0.01727 text-token cost. Its manual recovery and later 85-min deterministic replay are disclosed; zero residual simulator errors do not imply unattended operation or Reflection convergence. A separate historical-archive audit reproduces 96 cases within 0.0005 dB under pinned GNPY v2.12, while a schema-corrected 2.14.2 replay differs by 0.628 dB in GSNR on average. Equipment warnings persist. The ns-3 case demonstrates a second DT interface; the wireless modelling scope and validation are stated in its use-case section. Future work includes optical-equipment calibration, contemporaneous optical-wireless measurements, independent repeated workflow trials, and larger scenario spaces with explicit correctness and scalability metrics.

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